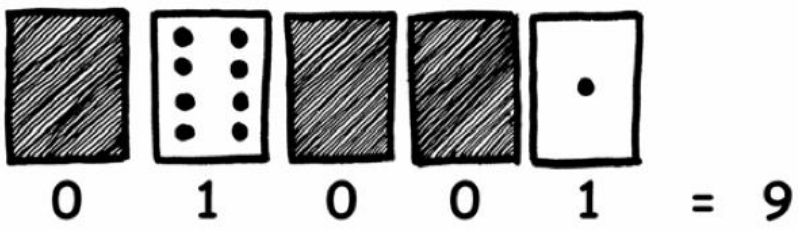
CSCI 1110
Assignment 2
100 points

Read the instructions carefully.
The goal for this assignment is:

- Practice the use of binary numbers

Data in computers is stored and transmitted as a series of zeros and ones. Let's practice how we represent words and numbers using just these two symbols.

1. (14 points) The binary system uses **zero** and **one** to represent whether a card is face up or not. **0** shows that a card is hidden, and **1** means that you can see the dots. For example:



a) Can you work out what the following numbers?

- 101011 = 75
- 100101 = 73
- 10101 = 21
- 0111 = 7
- 10111 = 23

b) What day of the month were you born? Write it in decimal and then in binary.

Day in decimal 09-01-93

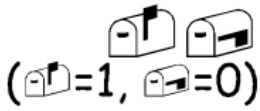
Day in binary -1001-1-10101101

2. (20 points) Try to work out these coded numbers:

☒☑☒☒☑ = 9
(☑=1, ☒=0)

⇧⇩⇧ = 5
(⇧=1, ⇩=0)

○○○○○ = 0
(⊙=1, ○=0)

 = 2
(=1, =0)

☹ = 0
(☺=1, ☹=0)

= 10
(=1, =0)

✚✚✖✚ = 13
(✚=1, ✖=0)

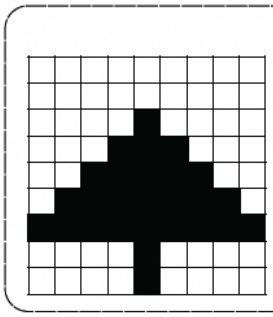
↻↻↻↻↻ = 17
(↻=1, ↺=0)

▲▼▲▼▼ = 20
(▲=1, ▼=0)

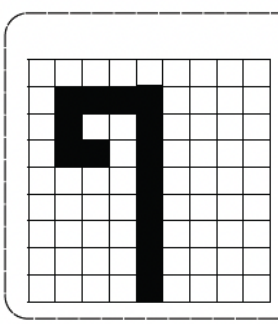
♠♠♠♠♠ = 31
(♠=1, ♣=0)

3. (20 points) Read the instructions for each part.

PART 1 Convert the bitmaps into binary using a "0" to represent a white pixel and a "1" to represent a black pixel.

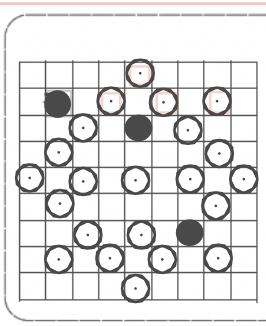


Binary
000000000
000000000
000010000
000111000
001111100
011111110
111111111
000010000
000010000

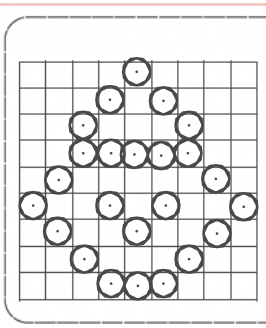


Binary
000000000
011110000
010010000
011010000
000010000
000010000
000010000
000010000
000010000

PART 2 Convert the binary into bitmaps. Use white to represent 0 and black to represent 1.



Binary
000010000
010101010
001010100
010000010
101010101
010000010
001010100
010101010
000010000



Binary
000010000
000101000
001000100
001111100
010000010
100101001
010010010
001000100
000111000

4. (18 points) Another interesting property of binary numbers is what happens when a zero is put on the right-hand side of the number. If we are working in base 10 (decimal), when you put a zero on the right-hand side of the number, it is multiplied by 10. For example, 9 becomes 90, 30 becomes 300.

a) But what happens when you put a 0 on the right of a binary number? **It multiplies the number by 2 if it's a normal binary number.**

b) What is the rule? **Adding a 0 to the right side of a binary number shifts the numbers to the left and the numbers into a new 2 to the power position, which just happened to multiply the number by 2.**

c) How many bits does a computer need to store all the characters at a keyboard? Do a little research if you don't know this answer. **At least 7 bits but 8 bits for the modern computer.**

5. (28 points) ASCII is like that secret code for computers. It stands for "American Standard Code for Information Interchange." In ASCII, each letter, number, and symbol are assigned a unique number.

So, when you type something on a computer, like "HELLO," behind the scenes, the computer converts each letter into its ASCII number. When you save a document or send a message, the computer stores or sends these numbers. Later, when you open the document or read the message, the computer changes the ASCII numbers back into the letters, and you can see the words again.

Can you crack the secret message? Use the **ASCII** table (https://www.asciitable.com) and find the secret message.

073 032 108 105 107 101 032 115 116 117 100 121 105 110 103 032 110 101 119 032 116 104 105 110 103 115 032 097 098 111 117 116 032 099 111 109 112 117 116 101 114 115 033

I like studying new things about computers!

Submission:

1. Submit to Canvas your answers as a **single** **PDF** file with the following name ***CS1110_LastName_FirstName_2.pdf***

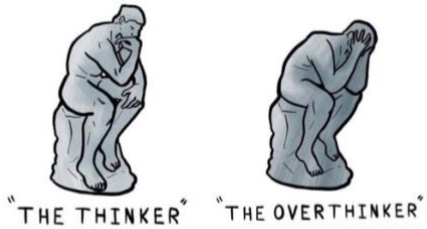
You can write your answers by hand or using a Word processor. For either way, make your answers clear and organized. If I cannot read or understand your answers I might give you the incorrect mark.

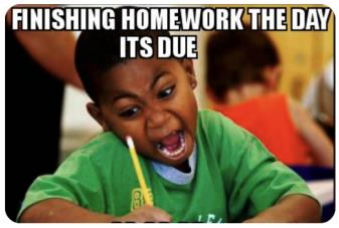
Make sure to check the **due date** for this activity on Canvas. **There is no late submission.** I do not accept submissions via email or as a comment on Canvas.

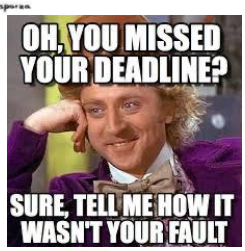
Unexpected situations might arise in life. Try to **submit your work before the due date** so you can have time to check for improvements.

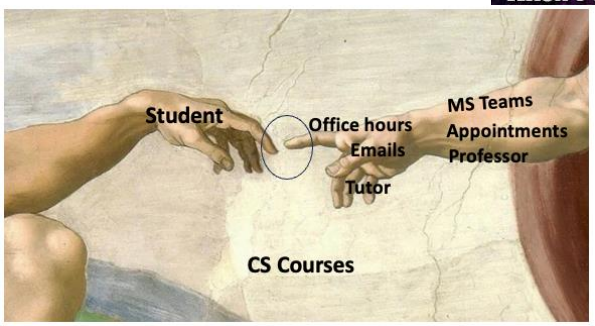
Make sure you are **submitting the correct files**. I will grade the files uploaded to Canvas.

Ensure that your work is original and created by you. It is YOUR responsibility to contact me if something from these instructions is not clear or ambiguous. Do not assume requirements. Do not overthink.


"THE THINKER"
"THE OVERTHINKER"


FINISHING HOMEWORK THE DAY
ITS DUE


OH, YOU MISSED
YOUR DEADLINE?
SURE, TELL ME HOW IT
WASN'T YOUR FAULT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